

1. A LIST OF STANDARD UNIVARIATE PROBABILITY DISTRIBUTIONS - PART 1

1.1. Discrete probability distributions and Discrete RVs.

Example 1.1 (Degenerate RV). Fix $c \in \mathbb{R}$. Say that X is degenerate at c if its distribution is given by the p.m.f.

$$f_X(x) = \mathbb{P}(X = x) = \begin{cases} 1, & \text{if } x = c \\ 0, & \text{otherwise.} \end{cases}$$

This is a discrete RV with support $S_X = \{c\}$. As computed earlier, $\mathbb{E}X = c$. We also have $\mathbb{E}X^n = c^n, \forall n \geq 1$ and $M_X(t) = e^{tc}, \forall t \in \mathbb{R}$. Note that $\text{Var}(X) = \mathbb{E}X^2 - (\mathbb{E}X)^2 = 0$.

Remark 1.2 (Bernoulli Trial). Suppose that a random experiment has exactly two outcomes, identified as a ‘success’ and a ‘failure’. For example, while tossing a coin, we may think of obtaining a head as a success and a tail as a failure. Here, the sample space is $\Omega = \{Success, Failure\}$. A single trial of such an experiment is referred to as a Bernoulli trial. In this case, $\text{Probability}(\{Success\}) = 1 - \text{Probability}(\{Failure\})$. If we define an RV $X : \Omega \rightarrow \mathbb{R}$ by $X(Success) = 1$ and $X(Failure) = 0$, then X is a discrete RV with p.m.f.

$$f_X(x) = \mathbb{P}(X = x) = \begin{cases} 1 - \text{Probability}(\{Success\}), & \text{if } x = 0, \\ \text{Probability}(\{Success\}), & \text{if } x = 1, \\ 0, & \text{otherwise.} \end{cases}$$

If $\text{Probability}(\{Success\}) = 0$ or $\text{Probability}(\{Failure\}) = 0$, then X is degenerate at 0 or 1, respectively. The case when $\text{Probability}(\{Success\}) \in (0, 1)$ is therefore of interest.

Example 1.3 (Bernoulli(p) RV). Let $p \in (0, 1)$. An RV X is said to follow Bernoulli(p) distribution or equivalently, X is a Bernoulli(p) RV if its distribution is given by the p.m.f.

$$f_X(x) = \begin{cases} 1 - p, & \text{if } x = 0, \\ p, & \text{if } x = 1, \\ 0, & \text{otherwise.} \end{cases}$$

In relation with the Bernoulli trial described above, p may be treated as the probability of success. Here, $\mathbb{E}X = p$, $\mathbb{E}X^2 = p$, $Var(X) = \mathbb{E}X^2 - (\mathbb{E}X)^2 = p - p^2 = p(1 - p)$, $M_X(t) = 1 - p + pe^t$, $t \in \mathbb{R}$. By standard arguments, we can establish the existence of these moments.

Notation 1.4. We may write $X \sim \text{Bernoulli}(p)$ to mean that X is a Bernoulli(p) RV. Similar notations shall be used for other RVs and their distributions.

Example 1.5 (Binomial(n, p) RV). Fix a positive integer n and let $p \in (0, 1)$. By the Binomial theorem, we have

$$1 = [p + (1 - p)]^n = \sum_{k=0}^n \binom{n}{k} p^k (1 - p)^{n-k}$$

and hence the function $f : \mathbb{R} \rightarrow [0, 1]$ given by

$$f(x) = \begin{cases} \binom{n}{x} p^x (1 - p)^{n-x}, & \text{if } x \in \{0, 1, \dots, n\}, \\ 0, & \text{otherwise.} \end{cases}$$

is a p.m.f.. An RV X is said to follow Binomial(n, p) distribution or equivalently, X is a Binomial(n, p) RV if its distribution is given by the above p.m.f.. Here,

$$\mathbb{E}X = \sum_{k=0}^n k \frac{n!}{k!(n-k)!} p^k (1 - p)^{n-k} = np \sum_{k=1}^n \binom{n-1}{k-1} p^{k-1} (1 - p)^{n-k} = np [p + (1 - p)]^{n-1} = np,$$

and

$$\mathbb{E}X(X-1) = \sum_{k=0}^n k(k-1) \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} = n(n-1)p^2 \sum_{k=2}^n \binom{n-2}{k-2} p^{k-2} (1-p)^{n-k} = n(n-1)p^2.$$

Then $\mathbb{E}X^2 = \mathbb{E}X(X-1) + \mathbb{E}X = n(n-1)p^2 + np$ and $Var(X) = \mathbb{E}X^2 - (\mathbb{E}X)^2 = n(n-1)p^2 + np - n^2p^2 = np(1 - p)$. Also

$$M_X(t) = \mathbb{E}e^{tX} = \sum_{k=0}^n e^{tk} \binom{n}{k} p^k (1 - p)^{n-k} = (1 - p + pe^t)^n, \forall t \in \mathbb{R}.$$

By standard arguments, we can establish the existence of these moments.

Note 1.6. Observe that Binomial($1, p$) distribution is the same as Bernoulli(p) distribution. We shall explore the connection between Binomial and Bernoulli distributions later in the course.

Remark 1.7 (Factorial moments). In the computation for $\mathbb{E}X^2$ for $X \sim \text{Binomial}(n, p)$, we first computed $\mathbb{E}X(X-1)$, which is easy to compute. It turns out that expectations of the form $\mathbb{E}X(X-1)$, $\mathbb{E}X(X-1)(X-2)$ etc. are often easy to compute for integer valued RVs X . We refer to such expectations as factorial moments of X .

Remark 1.8 (Symmetry of $\text{Binomial}(n, \frac{1}{2})$ distribution). Let $X \sim \text{Binomial}(n, p)$ and let $Y := n - X$. Since $M_X(t) = (1 - p + pe^t)^n, \forall t \in \mathbb{R}$, we have

$$M_Y(t) = \mathbb{E}e^{tY} = \mathbb{E}e^{t(n-X)} = e^{-nt}M_X(-t) = e^{nt}(1 - p + pe^{-t})^n = (p + (1 - p)e^t)^n.$$

Since MGFs determine the distribution, we conclude that $Y \sim \text{Binomial}(n, 1 - p)$. In particular, if $p = \frac{1}{2}$, then $Y = n - X \stackrel{d}{=} X \sim \text{Binomial}(n, \frac{1}{2})$. Rewriting the relation, we get $\frac{n}{2} - X \stackrel{d}{=} X - \frac{n}{2}$. Therefore, $X \sim \text{Binomial}(n, \frac{1}{2})$ is symmetric about $\frac{n}{2}$.

We now look at more examples of discrete RVs. Later in the course, we shall discuss their motivation through various random experiments.

Example 1.9 (Uniform RVs with support on a finite set). Consider a discrete RV X with support $S_X = \{x_1, x_2, \dots, x_n\}$ and p.m.f. $f_X : \mathbb{R} \rightarrow [0, 1]$ given by

$$f_X(x) = \begin{cases} \frac{1}{n}, & \text{if } x \in S_X, \\ 0, & \text{otherwise.} \end{cases}$$

We have

$$\mathbb{E}X = \frac{1}{n} \sum_{x \in S_X} x, \quad \mathbb{E}X^2 = \frac{1}{n} \sum_{x \in S_X} x^2, \quad M_X(t) = \mathbb{E}e^{tX} = \frac{1}{n} \sum_{x \in S_X} e^{tx}, \forall t \in \mathbb{R}$$

and hence $\text{Var}(X)$ can be computed by the formula $\mathbb{E}X^2 - (\mathbb{E}X)^2$. Observing that the support set is finite, we can establish the existence of these moments.

Example 1.10 (Poisson (λ) RV). Fix $\lambda > 0$. Note that $e^\lambda = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$ and hence the function $f : \mathbb{R} \rightarrow [0, 1]$ given by

$$f(x) = \begin{cases} e^{-\lambda} \frac{\lambda^x}{x!}, & \text{if } x \in \{0, 1, 2, \dots\}, \\ 0, & \text{otherwise.} \end{cases}$$

is a p.m.f.. An RV X is said to follow Poisson(λ) distribution or equivalently, X is a Poisson(λ) RV if its distribution is given by the above p.m.f.. Recall that we have already computed the following $\mathbb{E}X = \lambda$, $Var(X) = \lambda$ and $M_X(t) = e^{\lambda(e^t-1)}$, $\forall t \in \mathbb{R}$. As done for the case of Binomial(n, p) RVs, we can also compute the factorial moments. For example,

$$\mathbb{E}X(X-1) = \sum_{k=0}^{\infty} k(k-1)e^{-\lambda} \frac{\lambda^k}{k!} = \lambda^2 \sum_{k=2}^{\infty} e^{-\lambda} \frac{\lambda^{k-2}}{(k-2)!} = \lambda^2.$$

In fact, $\mathbb{E}X(X-1)\cdots(X-(n-1)) = \lambda^n$ for all $n \geq 1$.

Example 1.11 (Geometric (p) RV). Fix $p \in (0, 1)$. Note that $\sum_{k=0}^{\infty} p(1-p)^k = 1$ and hence the function $f : \mathbb{R} \rightarrow [0, 1]$ given by

$$f(x) = \begin{cases} p(1-p)^x, & \text{if } x \in \{0, 1, 2, \dots\}, \\ 0, & \text{otherwise.} \end{cases}$$

is a p.m.f.. An RV X is said to follow Geometric(p) distribution or equivalently, X is a Geometric(p) RV if its distribution is given by the above p.m.f.. Let us compute the MGF. Here,

$$M_X(t) = \mathbb{E}e^{tX} = \sum_{k=0}^{\infty} e^{tk} p(1-p)^k = \frac{p}{1 - (1-p)e^t},$$

for all t such that $0 < (1-p)e^t < 1$ or equivalently, $t < \ln\left(\frac{1}{1-p}\right)$. Looking at the derivatives of M_X and evaluating at $t = 0$, we have $\mathbb{E}X = \frac{1-p}{p}$ and $Var(X) = \frac{1-p}{p^2}$.

1.2. Continuous probability distributions and Continuous RVs.

Example 1.12 (Uniform(a, b) RV). Fix $a, b \in \mathbb{R}$ with $a < b$. An RV X is said to follow Uniform(a, b) distribution or equivalently, X is a Uniform(a, b) RV if its distribution is given

by the p.d.f.

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & \text{if } x \in (a, b), \\ 0, & \text{otherwise.} \end{cases}$$

We have

$$\mathbb{E}X = \int_a^b \frac{x}{b-a} dx = \frac{b^2 - a^2}{2(b-a)} = \frac{a+b}{2}, \quad \mathbb{E}X^2 = \int_a^b \frac{x^2}{b-a} dx = \frac{b^3 - a^3}{3(b-a)} = \frac{a^2 + ab + b^2}{3}$$

and hence $Var(X)$ can be computed by the formula $\mathbb{E}X^2 - (\mathbb{E}X)^2$. The MGF is given by

$$\mathbb{E}e^{tX} = \int_a^b \frac{e^{tx}}{b-a} dx = \begin{cases} \frac{e^{tb} - e^{ta}}{t(b-a)}, & \text{if } t \neq 0, \\ 1, & \text{if } t = 0. \end{cases}$$

By standard arguments, we can establish the existence of these moments. Further, observe that $f_X(\frac{a+b}{2} - x) = f_X(\frac{a+b}{2} + x), \forall x \in \mathbb{R}$. Here, we conclude that X is symmetric about its mean $\frac{a+b}{2}$.

Example 1.13 (Cauchy(μ, θ) RV). Let $\theta > 0$ and $\mu \in \mathbb{R}$. An RV X is said to follow Cauchy(μ, θ) distribution if its distribution is given by the p.d.f.

$$f_X(x) = \frac{\theta}{\pi} \frac{1}{\theta^2 + (x - \mu)^2}, \forall x \in \mathbb{R}.$$

The fact that f_X is a p.d.f. is easy to check. Set $y = \frac{x-\mu}{\theta}$ and observe that

$$\int_{-\infty}^{\infty} f_X(x) dx = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{1}{1+y^2} dy = \frac{2}{\pi} \int_0^{\infty} \frac{1}{1+y^2} dy = \frac{2}{\pi} \tan^{-1}(y) \Big|_0^{\infty} = 1.$$

We have seen earlier that $\mathbb{E}X$ and the MGF do not exist for this distribution. In the general setting, note that $\frac{X-\mu}{\theta} \sim \text{Cauchy}(0, 1)$ and by a similar argument, we can show that $\mathbb{E}X$ and MGF do not exist. Moreover, $f_X(\mu + x) = f_X(\mu - x), \forall x \in \mathbb{R}$. Hence, we conclude that X is symmetric about μ .

Example 1.14 (Exponential(λ) RV). Let $\lambda > 0$. Note that $\int_0^\infty \exp(-\frac{x}{\lambda}) dx = \lambda$ and hence the function $f : \mathbb{R} \rightarrow [0, \infty)$ given by

$$f(x) = \begin{cases} \frac{1}{\lambda} \exp(-\frac{x}{\lambda}), & \text{if } x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

is a p.d.f.. An RV X is said to follow Exponential(λ) distribution or equivalently, X is an Exponential(λ) RV if its distribution is given by the above p.d.f.. We have

$$\mathbb{E}X^n = \lambda^n n!, \quad \text{Var}(X) = \lambda^2, \quad M_X(t) = (1 - \lambda t)^{-1}, \forall t < \frac{1}{\lambda}.$$

By standard arguments, we can establish the existence of these moments.

Definition 1.15 (Gamma function). Recall that the integral $\int_0^\infty x^{\alpha-1} e^{-x} dx$ exists if and only if $\alpha > 0$. On $(0, \infty)$, consider the function $\alpha \mapsto \int_0^\infty x^{\alpha-1} e^{-x} dx$. It is called the Gamma function and the value at any $\alpha > 0$ is denoted by $\Gamma(\alpha)$.

Remark 1.16. We recall some important properties of the Gamma function.

- (a) For $\alpha > 0$, we have $\Gamma(\alpha) > 0$.
- (b) $\Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1)$, if $\alpha > 1$.
- (c) $\Gamma(1) = \int_0^\infty e^{-x} dx = 1$ and hence using (b), $\Gamma(n) = (n - 1)!$ for all positive integers n .
- (d) $\Gamma(\frac{1}{2}) = \int_0^\infty \frac{1}{\sqrt{x}} e^{-x} dx = \sqrt{\pi}$. Putting $x = \frac{y^2}{2}$, this relation may be rewritten as

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{2} \int_0^\infty \exp\left(-\frac{y^2}{2}\right) dy = \sqrt{\pi}.$$

- (e) Fix $\beta > 0$. Putting $x = \frac{y}{\beta}$, in the integral for $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$, we get $\Gamma(\alpha) = \int_0^\infty y^{\alpha-1} \beta^{-\alpha} \exp(-\frac{y}{\beta}) dy$.

Example 1.17 (Gamma(α, β) RV). Fix $\alpha > 0, \beta > 0$. By the properties of the Gamma function described above, the function $f : \mathbb{R} \rightarrow [0, \infty)$ defined by

$$f(x) = \begin{cases} \frac{1}{\Gamma(\alpha)} x^{\alpha-1} \beta^{-\alpha} \exp(-\frac{x}{\beta}), & \text{if } x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

is a p.d.f.. An RV X is said to follow Gamma(α, β) distribution or equivalently, X is a Gamma(α, β) RV if its distribution is given by the above p.d.f.. Note that for $\alpha = 1$, we get back the p.d.f. for an Exponential(β) RV (see Example 1.14), i.e. Gamma(1, β) distribution is the same as Exponential(β) distribution. For general $\alpha > 0, \beta > 0$, we have

$$\mathbb{E}X = \alpha\beta, \quad \text{Var}(X) = \alpha\beta^2, \quad M_X(t) = (1 - \beta t)^{-\alpha}, \forall t < \frac{1}{\beta}.$$

By standard arguments, we can establish the existence of these moments.

Example 1.18 (Normal(μ, σ^2) RV). Fix $\mu \in \mathbb{R}, \sigma > 0$. Note that $\Gamma(\frac{1}{2}) = \int_0^\infty \frac{1}{\sqrt{t}} e^{-t} dt = \sqrt{\pi}$ (see Remark 1.16). Putting $t = \frac{y^2}{2}$ and after suitable manipulation, we have $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty \exp\left(-\frac{y^2}{2}\right) dy = 1$. Putting $y = \frac{1}{\sigma}(x - \mu)$ (equivalently, $x = \sigma y + \mu$), we have

$$\frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^\infty \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) dx = 1.$$

Therefore, the function $f : \mathbb{R} \rightarrow [0, \infty)$ defined by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right), \forall x \in \mathbb{R}$$

is a p.d.f.. An RV X is said to follow Normal(μ, σ^2) distribution or equivalently, X is a Normal(μ, σ^2) RV, denoted by $X \sim N(\mu, \sigma^2)$ if its distribution is given by the above p.d.f.. If $X \sim N(\mu, \sigma^2)$, from our above discussion we conclude that $Y = \frac{X - \mu}{\sigma} \sim N(0, 1)$. Now,

$$\begin{aligned} M_Y(t) &= \mathbb{E}e^{tY} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty e^{ty} \exp\left(-\frac{y^2}{2}\right) dy \\ &= \exp\left(\frac{t^2}{2}\right) \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty \exp\left(-\frac{(y - t)^2}{2}\right) dy \end{aligned}$$

$$= \exp\left(\frac{t^2}{2}\right), \forall t \in \mathbb{R}.$$

In particular, $\psi_Y(t) = \ln M_Y(t) = \frac{t^2}{2}, \forall t \in \mathbb{R}$ with $\psi'(t) = t, \psi''(t) = 1, \forall t \in \mathbb{R}$. Evaluating at $t = 0$, we conclude that $\mathbb{E}Y = 0$ and $\text{Var}(Y) = 1$. But $X = \sigma Y + \mu$ and hence $\mathbb{E}X = \mu, \text{Var}(X) = \sigma^2$. This yields the interpretation of the parameters μ and σ in the distribution of X . Further, $M_X(t) = \mathbb{E}e^{tX} = \mathbb{E}e^{t(\sigma Y + \mu)} = e^{\mu t} M_Y(\sigma t) = \exp(\mu t + \frac{1}{2}\sigma^2 t^2), \forall t \in \mathbb{R}$.

Definition 1.19 (Standard Normal RV). We say X is a Standard Normal RV if $X \sim N(0, 1)$, i.e. $\mathbb{E}X = 0$ and $\text{Var}(X) = 1$.

Notation 1.20. Normal RVs are also referred to as Gaussian RVs and Normal distribution as Gaussian distribution.

Remark 1.21 (Symmetry of Gaussian Distribution). If $X \sim N(\mu, \sigma^2)$, note that $f_X(\mu + x) = f_X(\mu - x), \forall x \in \mathbb{R}$. Hence, we conclude that X is symmetric about its mean μ .

Remark 1.22 (Moments of a Standard Normal RV). Let $X \sim N(0, 1)$. Then X is symmetric about 0 and consequently, $\mathbb{E}X^n = 0$ for all odd positive integers n . If n is an even positive integer, then $n = 2m$ for some positive integer m and

$$\begin{aligned} \mathbb{E}X^n &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x^n \exp\left(-\frac{x^2}{2}\right) dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x^{2m} \exp\left(-\frac{x^2}{2}\right) dx \\ &= \frac{\sqrt{2}}{\sqrt{\pi}} \int_0^{\infty} x^{2m} \exp\left(-\frac{x^2}{2}\right) dx \\ &= \frac{2^m}{\sqrt{\pi}} \int_0^{\infty} y^{m-\frac{1}{2}} \exp(-y) dy, \text{ (putting } y = \frac{x^2}{2}\text{)} \\ &= \frac{2^m}{\sqrt{\pi}} \Gamma\left(m + \frac{1}{2}\right) \\ &= 2^m \left(m - \frac{1}{2}\right) \times \cdots \times \frac{3}{2} \times \frac{1}{2} \\ &= (2m - 1) \times \cdots \times 3 \times 1 =: (2m - 1)!!, \end{aligned}$$

where we have used the properties of the Gamma function. In particular, $\mathbb{E}X^4 = 3$.

Definition 1.23 (Beta function). Recall that the integral $\int_0^1 x^{\alpha-1}(1-x)^{\beta-1} dx$ exists if and only if $\alpha > 0$ and $\beta > 0$. On $(0, \infty) \times (0, \infty)$, consider the function $(\alpha, \beta) \mapsto \int_0^1 x^{\alpha-1}(1-x)^{\beta-1} dx$. It is called the Beta function and the value at any (α, β) is denoted by $B(\alpha, \beta)$.

Remark 1.24. Note that for $\alpha > 0, \beta > 0$, we have $B(\alpha, \beta) > 0$ and $B(\alpha, \beta) = B(\beta, \alpha)$. Moreover,

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}.$$

Example 1.25 (Beta(α, β) RV). Fix $\alpha > 0, \beta > 0$. By the properties of the Beta function described above, the function $f : \mathbb{R} \rightarrow [0, \infty)$ defined by

$$f(x) = \begin{cases} \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}, & \text{if } x \in (0, 1) \\ 0, & \text{otherwise.} \end{cases}$$

is a p.d.f.. An RV X is said to follow Beta(α, β) distribution or equivalently, X is a Beta(α, β) RV if its distribution is given by the above p.d.f.. If $\alpha = \beta$, then $f(1-x) = f(x), \forall x \in \mathbb{R}$ and hence $X \stackrel{d}{=} 1 - X$. Then, $X - \frac{1}{2} \stackrel{d}{=} \frac{1}{2} - X$, i.e., X is symmetric about $\frac{1}{2}$. For all $\alpha, \beta, r > 0$, we have

$$\mathbb{E}X^r = \frac{1}{B(\alpha, \beta)} \int_0^1 x^{\alpha+r-1} (1-x)^{\beta-1} dx = \frac{B(\alpha+r, \beta)}{B(\alpha, \beta)}$$

and in particular,

$$\mathbb{E}X = \frac{B(\alpha+1, \beta)}{B(\alpha, \beta)} = \frac{\Gamma(\alpha+1)\Gamma(\beta)}{\Gamma(\alpha+\beta+1)} \times \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} = \frac{\alpha}{\alpha+\beta}$$

and

$$\mathbb{E}X^2 = \frac{B(\alpha+2, \beta)}{B(\alpha, \beta)} = \frac{\Gamma(\alpha+2)\Gamma(\beta)}{\Gamma(\alpha+\beta+2)} \times \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} = \frac{(\alpha+1)\alpha}{(\alpha+\beta+1)(\alpha+\beta)}.$$

Then

$$\text{Var}(X) = \mathbb{E}X^2 - (\mathbb{E}X)^2 = \frac{\alpha\beta}{(\alpha+\beta+1)(\alpha+\beta)^2}.$$

By standard arguments, we can establish the existence of these moments.